FEMINIST REPLICATION PAPER

Folbre, N., & Smith, K. (2017).
The wages of care: Bargaining power, earnings and inequality.

Introduction

The paper written by Nancy Folbre and Kristin Smith, one of the profound feminist economists addresses the care-work with the gender relations of power. Also, the authors reiterate that gender doesn't define who gets to do the care work like eldercare, childcare, nursing but the social norms of the society decide about the type of work women can take part in and how undervalued the work is in the economy. We can relate this paper with the findings of paper "Valuing Work" by "Devaki Jain" that explains that "Feminism argues that women don't have any issues with doing the care-work (eldercare, child care) but where are the men and why is it women's work and if its women's work, why is care-work paid very low wages". For example: suppose if men were the ones dominating these care professions , would they have accepted it; getting low pennies, treated with no respect for their work, complete invisibility of their work in GDP counting. No way!. Feminists aren't just criticizing the low wage-pay but the structure that assigns non-market value to labor that keeps society running. The paper also talks about bargaining power, earnings and inequality are determining how gender power relations play out in the market to determine the wages of care-work.

Data

The data used for this replication exercise and analysis is only from the **Current Population Survey, ASEC-2015**. The raw dataset was obtained from **IPUMS CPS** website. Downloading the dataset required identifying the variables with which we wanted to replicate the results in our study.

Methodology

- The dataset downloaded from IPUMS CPS website was raw and uncleaned.
- The replication of results required filtering of the dataset, variable selection, and calculating hourly wages.
- All the data cleaning, manipulation and statistical results were performed using R software.
- The data was filtered according to the criteria:
    - Workers currently employed and salaried workers.
    - Workers with positive wage income (incwage>0).

    - Workers with invalid or 000 codes were excluded as they were not included in the universe or they were invalid.
    - Workers with age 15-66 were taken for replication.

- The total number of observations in the dataset was **77,584**. Due to data constraints, I was able to replicate **77,358** observations.
- The independent variable in the study is **Care Industry and Other Industry**. Control variables include **age, education level**. Dummy variables include **marital status, number of children in the household, race/ethnicity**. The dependent variable in the study is Log Hourly wage.
- Creating care and other industry sectors with the help of codes provided by the author.
- The variables taken were sex, age, race, education, marital status, hours worked, number of children in household, hispanic origin, employment status, class of worker, weeks worked last year, wage and salary income. The variables were changed from initial variables because data bias was showing, therefore, used alternative variables.

Results

Table1: Type of Employer in Care and Other Industries

Table 1: Type of Employer in Care and Other Industries (Replication)

Row Label	Care Industries	Other
Total	24.7	75.3
Women	38.5	61.5
Men	11.5	88.5
Private, for-profit	19.8	80.2
Women	32.5	67.5
Men	8.2	91.8
Public	48.1	51.9
Women	62.3	37.7
Men	29.9	70.1

Interpretation:

The table illustrates that the care industry is dominated by women and this trend can be seen specifically in private for profit and public sectors, where the public sector has a smaller share than the private sector.

Table 2: Annual Earnings at the 10th, 50th, and 90th Percentiles by Type of Industry

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Row Label	P10	P50 (Median)	P90	P10/P50	P50/P90
Care Industries	$13,000	$40,784	$89,569	31.9	45.5
Women	$12,235	$36,706	$80,549	33.3	45.6
Men	$16,314	$50,447	$122,353	32.3	41.2
Other Industry	$12,235	$38,745	$101,961	31.6	38.0
Women	$9,455	$30,588	$81,569	30.9	37.5
Men	$15,294	$45,882	$112,157	33.3	40.9

Interpretation:

The table compares the annual earnings between the care industry and other industries mainly focusing on low earners, medium earners, high earners and the ratios measuring the wage inequality within each group. The median earnings of the care industry are slightly more than other industries. Median earnings for men in the care industry is higher than in other industries and the same trend is seen in women. The high earners in other industries earn more than the workers in care industries which indicates that the care sector may have very few high paying jobs. The wage gap between bottom 10 and median is nearly identical in both sectors but the wage gap between median and top 10 is larger in the care industry as compared to other industries.

Table 3 : Characteristics of Workers by Type of Industry and Gender

Table 3: Characteristics of Workers by Type of Industry and Gender (Replication)

Characteristic	Care Industries Women	Care Industries Men	Other Industry Women	Other Industry Men
Age (mean)	42.0	42.2	39.9	40.3
Less than high school	3.3	3.4	10.3	11.9
High school degree	16.4	13.3	27.7	31.1
Some college	31.0	22.7	31.7	27.9
College graduate	26.9	26.3	21.9	20.2
Master's degree	17.8	20.0	6.4	6.6
Ph.D. or Professional degree	4.6	14.3	2.1	2.2
Percent married	60.6	64.5	49.3	60.5
Percent with children	59.9	51.4	52.0	51.1
White, non-Hispanic	67.4	65.9	60.5	62.4
Black, non-Hispanic	13.3	12.2	11.7	9.4
Other, non-Hispanic	6.6	10.0	8.6	7.4
Hispanic	12.6	11.9	19.2	20.8

Interpretation:

Workers in the care industry tend to be slightly older than workers in the other industries. Men in both the sectors have a higher rate of marriage as compared to women. But, women in care industries are slightly more likely to be married than the other industries. The care industry workforce are more highly educated as compared to other industry workforce at the graduate level. We can see distinct over-representation of married women and mothers, compared to other industry sectors.

Table 5: Ordinary Least Squares Model Predicting Natural Log of Hourly Earnings

	Model I	Model II	Model III
Care work industry	-0.12***	-0.07***	-0.11***
	(0.01)	(0.01)	(0.01)
Manager	0.77***	0.48***	0.43***
	(0.01)	(0.01)	(0.01)
Professional	0.67***	0.36***	0.30***
	(0.01)	(0.01)	(0.01)
Private, for-profit	0.02*	0.14***	2.78***
	(0.01)	(0.01)	(0.02)
Female		-0.22***	-0.25***
		(0.01)	(0.01)
Age		0.01***	0.01***
		(0.00)	(0.00)
High school degree		0.28***	0.29***
		(0.02)	(0.02)
Some college		0.37***	0.41***
		(0.02)	(0.02)
College degree		0.65***	0.69***
		(0.02)	(0.02)
Master's degree		0.78***	0.83***
		(0.02)	(0.02)
Ph.D./ Professional degree		0.97***	1.02***
		(0.03)	(0.03)
Married		0.10***	0.13***
		(0.01)	(0.01)
Children		0.12***	0.12***
		(0.01)	(0.01)
Black, not Hispanic		0.17	0.16
		(0.11)	(0.11)
Other, not Hispanic		0.27**	0.25*
		(0.10)	(0.10)
Hispanic		0.07**	0.08***
		(0.02)	(0.02)
Public sector			2.80***
			(0.02)
R2	0.082	0.154	0.198
Num.Obs.	77372	77358	77358

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Interpretation:

The table illustrates the results of three separate OLS regression models that analyze the factors that influence workers log hourly earnings. Workers in the care industry results in more wage penalty across all the groups which is larger for the pooled sample(model 1) and women (model 3) and less severe for men. And also in the pooled sample, the coefficient is very small, but for private- for profit shows a large premium for women indicating high return for women in certain for-profit care jobs.

Reflection

The overall analysis of the replicated table matches with the original paper by Nancy Folbre and Kristin Smith - "**The wages of care: Bargaining power, earnings and inequality",** the number of observations in the original table was 77, 584 but the replicated tables observation is 77, 358. This difference is the result of non-availability of data under private: not profit, only private: for-profit employment exists which results in little bias in the results and the figures in the table are a slight mismatch but the analysis, the conclusion remains the same. The replication confirms the demographic trends like age, marital status of the paper. The most important analysis is the workforce in the care industry is older, female-dominated and highly educated as compared to other industry workforce. The replication was overall successful and it provided many lessons like accurately replicating the sample filtering and variable selection from raw IPUMS CPS dataset.